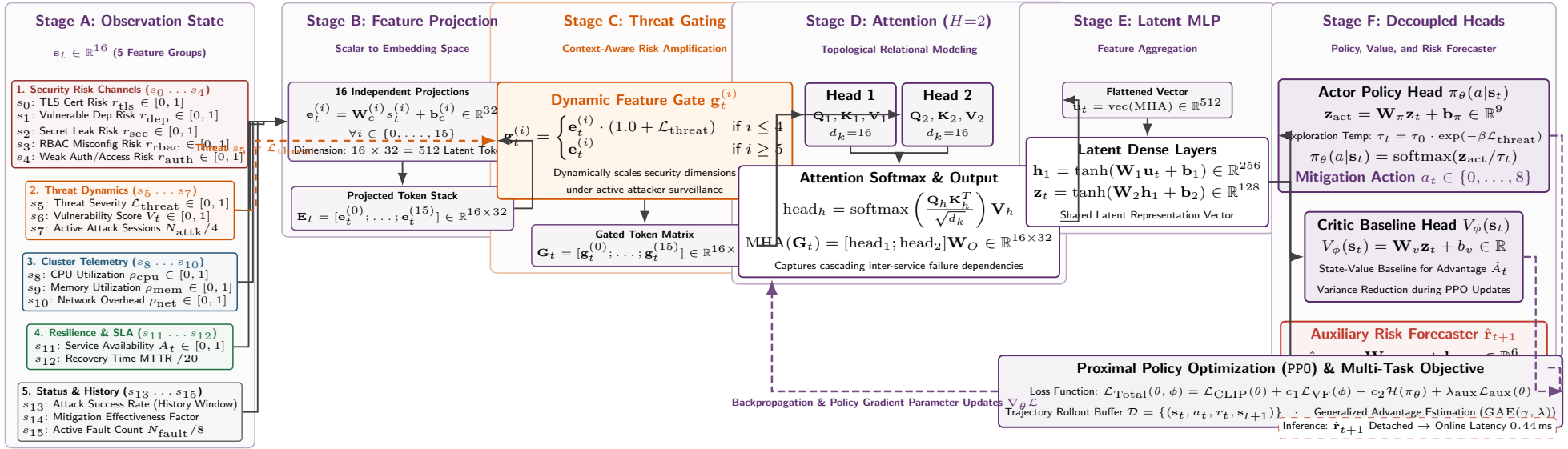

Stage A: Observation State
$s_t \in \mathbb{R}^{16}$ (5 Feature Groups)
1. Security Risk Channels ($s_0 \dots s_4$)
s_0: TLS Cert Risk $r_{\text{tls}} \in [0,1]$
s_1: Vulnerable Dep Risk $r_{\text{dep}} \in [0,1]$
s_2: Secret Leak Risk $r_{\text{sec}} \in [0,1]$
s_3: RBAC Misconfig Risk $r_{\text{rbac}} \in [0,1]$
s_4: Weak Auth/Access Risk $r_{\text{auth}} \in [0,1]$
2. Threat Dynamics ($s_5 \dots s_7$)
s_5: Threat Severity $\mathcal{L}_{\text{threat}} \in [0,1]$
s_6: Vulnerability Score $V_t \in [0,1]$
s_7: Active Attack Sessions $N_{\text{attk}}/4$
3. Cluster Telemetry ($s_8 \dots s_{10}$)
s_8: CPU Utilization $\rho_{\text{cpu}} \in [0,1]$
s_9: Memory Utilization $\rho_{\text{mem}} \in [0,1]$
s_{10}: Network Overhead $\rho_{\text{net}} \in [0,1]$
4. Resilience & SLA ($s_{11} \dots s_{12}$)
s_{11}: Service Availability $A_t \in [0,1]$
s_{12}: Recovery Time MTTR $/20$
5. Status & History ($s_{13} \dots s_{15}$)
s_{13}: Attack Success Rate (History Window)
s_{14}: Mitigation Effectiveness Factor
s_{15}: Active Fault Count $N_{\text{fault}}/8$
Threat $s_5 = \mathcal{L}_{\text{threat}}$
Stage B: Feature Projection
Scalar to Embedding Space
16 Independent Projections
$\mathbf{e}_t^{(i)} = \mathbf{W}_e^{(i)} s_t^{(i)} + \mathbf{b}_e^{(i)} \in \mathbb{R}^{32}$
$\forall i \in \{0, \dots, 15\}$
Dimension: $16 \times 32 = 512$ Latent Tok
Projected Token Stack
$\mathbf{E}_t = [\mathbf{e}_t^{(0)}; \dots; \mathbf{e}_t^{(15)}] \in \mathbb{R}^{16\times 32}$
Stage C: Threat Gating
Context-Aware Risk Amplification
Dynamic Feature Gate $\mathbf{g}_t^{(i)}$
$\mathbf{g}_t^{(i)} = \begin{cases} \mathbf{e}_t^{(i)} \cdot (1.0 + \mathcal{L}_{\text{threat}}) & \text{if } i \le 4 \\ \mathbf{e}_t^{(i)} & \text{if } i \ge 5 \end{cases}$
Dynamically scales security dimensions
under active attacker surveillance
Gated Token Matrix
$\mathbf{G}_t = [\mathbf{g}_t^{(0)}; \dots; \mathbf{g}_t^{(15)}] \in \mathbb{R}^{16\times$
Stage D: Attention (H=2)
Topological Relational Modeling
Head 1
$\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1$
d_k=16
Head 2
$\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2$
d_k=16
Attention Softmax & Output
$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}}\right) \mathbf{V}_h$
$\text{MHA}(\mathbf{G}_t) = [\text{head}_1; \text{head}_2]\mathbf{W}_O \in \mathbb{R}^{16\times 32}$
Captures cascading inter-service failure dependencies
Stage E: Latent MLP
Feature Aggregation
Flattened Vector
$\mathbf{u}_t = \text{vec}(\text{MHA}) \in \mathbb{R}^{512}$
Latent Dense Layers
$\mathbf{h}_1 = \tanh(\mathbf{W}_1 \mathbf{u}_t + \mathbf{b}_1) \in \mathbb{R}^{256}$
$\mathbf{z}_t = \tanh(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2) \in \mathbb{R}^{128}$
Shared Latent Representation Vector
Stage F: Decoupled Heads
Policy, Value, and Risk Forecaster
Actor Policy Head $\pi_\theta(a|\mathbf{s}_t)$
$\mathbf{z}_{\text{act}} = \mathbf{W}_\pi \mathbf{z}_t + \mathbf{b}_\pi \in \mathbb{R}^9$
Exploration Temp: $\tau_t = \tau_0 \cdot \exp(-\beta \mathcal{L}_{\text{threat}})$
$\pi_\theta(a|\mathbf{s}_t) = \text{softmax}(\mathbf{z}_{\text{act}}/\tau_t)$
Mitigation Action $a_t \in \{0, \dots, 8\}$
Critic Baseline Head $V_\phi(\mathbf{s}_t)$
$V_\phi(\mathbf{s}_t) = \mathbf{W}_v \mathbf{z}_t + b_v \in \mathbb{R}$
State-Value Baseline for Advantage $\hat{A}_t$
Variance Reduction during PPO Updates
Auxiliary Risk Forecaster $\hat{\mathbf{r}}_{t+1}$
Proximal Policy Optimization (PPO) & Multi-Task Objective
Loss Function: $\mathcal{L}_{\text{Total}}(\theta, \phi) = \mathcal{L}_{\text{CLIP}}(\theta) + c_1 \mathcal{L}_{\text{VF}}(\phi) - c_2 \mathcal{H}(\pi_\theta) + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}(\theta)$
Trajectory Rollout Buffer $\mathcal{D} = \{(\mathbf{s}_t, a_t, r_t, \mathbf{s}_{t+1})\}$
Generalized Advantage Estimation (GAE(γ, λ))
Inference: $\hat{\mathbf{r}}_{t+1}$ Detached → Online Latency 0.44 ms
Backpropagation & Policy Gradient Parameter Updates $\nabla_\theta \mathcal{L}$